

Approaches to Parts and Wholes in Semantics

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ESSLLI 2025

Advanced Course

Abstract

The course will give an overview of extensional and non-extensional mereological theories of parts and wholes, their historical predecessors and recent and new applications to natural language. It will discuss empirical and conceptual challenges for extensional mereological theories (in the tradition of Link 1983) as well as various approaches that makes use of a notion of an integrated whole (Langacker, Moltmann, Grimm, Filip, and others) or other notions of unity (Priest). The course will discuss key topics relating to the mass-count distinction and the various attempts by philosophers and linguists of clarifying the notion unity (or of being a single thing) that appears to be at the center of that distinction. It will also discuss other topics where the issue of unity arises, such as apparent polysemies (of nouns like 'book' and 'lunch'). The course aims to build a bridge between philosophical and linguistic traditions regarding mereology.

Motivation and description

It is generally agreed that part structures and notion of a whole or unity play an important role in the semantics of natural language. For example, mereology is standardly used for the semantics of plurals and mass nouns, and the notion of unity or being a single thing appears to be at the core of the content of the mass-count distinction. For a long time, the standard approach to part-whole relations and the mass-count distinction has been that of extensional mereology, largely due to the influential work of Link (1983). That is, part-whole structures consist in a partial order among entities that is closed under sum formation. The notion of a whole or having unity has usually been understood as that of being an atom (relative to a set or concept or specific part relation). There is an alternative tradition, though, on which a notion of boundary, structure, form, integrated whole or unifying function plays an important role for part-whole structures as well, especially for the notion of unity or being a single thing. Such notions have been argued to constitute the content of the mass-count distinction itself (Langacker, Jackendoff, Moltmann). A lot of recent empirical research has shown that conditions of integrity govern the choice of mass or count with nominalizations of various

sorts (Barner, Grimm, Filip, Sutton). Extensional mereological theories not only seem to impoverished to account for a range of linguistic facts; they also face various conceptual challenges. For example, they require distinguishing three different part relations, corresponding to the domains of singular count, mass, and plural nouns, without those being grounded in ontology itself (rather than being associated with syntactic category). The notion of an atom, as a construal of the notion of unity or a single thing is far from unproblematic, since atomicity itself has little to do with countability as such. But integrity-based approaches to part-whole structure also face challenges. There is the general issue of how the notion of an integrated whole is to be defined in the first place. Moreover, integrity in any worldly or cognitive sense seems absent when talking about ‘quantities’, ‘amounts’, ‘sums’, ‘loose collections’, ‘beings’ and alike, using singular count nouns.

The notion of unity pertains not only to the mass-count distinction, but to a broader range of issues, some of which have received considerably more attention in philosophy than linguistics, such as the unity of form and matter, the unity of the proposition, and the issue whether there can be a single thing, say a variable object, that apparently polysemous nouns stand for (*book, lunch, college*).

The course will give an overview of extensional and integrity-based theories of part-whole structure and critically discuss various unifying notions of integrity or function, including Aristotelian notions of form and structure, notions of boundary in the cognitive semantic literature, and the notion of integrated whole as part of situated part structures of my own previous work. It will discuss applications of the notion of integrated whole as part of situated part structures (the semantics of *whole, entire, wholly, completely, together, time(s)*) as well as recent applications of such notions to mass-count related phenomena by Grimm, Filip and Sutton and others. It will discuss approaches on which unity is considered a primitive, such as that of Link’s 1983 proposal, Priest’s account of unity in terms of a primitive notion of a *gluon*, McKay’s distinction between singular and mass reference, as well as some recent proposals of my own. In particular, it will address conceptual challenges for such views, as well as their empirical adequacy with respect to the relevant range of semantic issues. Finally, the course will address the question whether there can be a single notion of unity applying to the range of semantic and philosophical issues for which such a notion has been invoked. The course will aim to highlight the relevance of philosophical traditions for current semantic debates and to show the fruitfulness of employing lesser known notions from philosophical ontology for semantic analysis.

Tentative schedule

Session 1:

Extensional mereology, its applications, and its challenges

Session 2:

Unifying notions of integrity and function: historical views (Aristotle, Langacker, Jackendoff, Simons, and others)

Session 3:

Recent and new research using notions of integrated whole 1:

The theory of situated part structures and the semantics of *whole*, *together*, *time*, etc,

Session 4:

Recent and new research using notions of integrated whole 2:

Choices of mass and count for nominalizations, apparent polysemies

Session 5:

Unity as a primitive: proposals and their applications, prospects and challenges.

References

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Carrara, M., A. Arapinis, and F. Moltmann (eds.) (2017): *Unity and Plurality. Logic, Philosophy, and Semantics*. Oxford UP, Oxford.

Champollion, L. and M. Krifka (2017): *Mereology*. In P. Dekker and M. Aloni (eds): *Cambridge Handbook of Semantics*. Cambridge UP, Cambridge.

Chierchia, G. (1998): 'Plurality of Mass Nouns and the Notion of 'Semantic Parameter''. In S. Rothstein (ed.): *Events and Grammar*, Kluwer, Dordrecht, 53–103.

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Doetjes J. (2012): 'Count/Mass Distinctions across Languages'. In C. Maienborn, K. v. Heusinger, and P. Portner (eds.): *Semantics: an International Handbook of Natural Language Meaning, Part III*. De Gruyter. Berlin, 2559-2580.

Filip, H. (2021): *Countability in Natural Language*. Cambridge UP, Cambridge.

Grimm, S. (2018): 'Grammatical Number and the Scale of Individuation'. *Language* 94.3, 527-574.

Jackendoff, R. (1991): 'Parts and Boundaries'. *Cognition* 41, 9-45.

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- Koslicki, K. (2008): *The Structure of Objects*, Oxford UP, Oxford.
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- (2005): 'Part Structures in Situations: The Semantics of *Individual* and *Whole*'. *Linguistics and Philosophy* 28(5), 599-641
- (2021): 'Levels of Ontology and Natural Language: The Case of the Ontology of Parts and Wholes'. In J. Miller (ed.): *The Language of Ontology*. Oxford University Press, Oxford.
- (to appear): 'Metaphysics First or Language First? The Notion of a Single Object'. In R. Gaskin (ed.): *The Question of Linguistic Idealism*. Oxford UP, Oxford
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- Rothstein, S. (2017): *Semantics for Counting and Measuring*. Cambridge, Cambridge UP, Cambridge.
- Pelletier, F. and L. Schubert (2012): 'Mass Expressions'. In F. Guenther and D. Gabbay (eds.): *Handbook of Philosophical Logic*, 2nd edition. Vol. 10. Kluwer, Dordrecht, 249– 336.
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Course Outline

Session 1:

Ontological and Semantic Issues about Part-Whole Structure

Linguistic intuitions and part-whole structure

Three views of part-whole structure:

- Extensional mereology
- Mereotopology and hylomorphism
- The priority of the whole

Session 2:

Extensional mereology: motivations, applications, and problems

Reading:

Link, G. (1983): 'The Logical Analysis of Plurals and Mass Nouns'. In R. Baeuerle et al. (eds.): *Semantics from Different Points of View*. Springer, Berlin, 302-323.

Varzi, A. (2016): 'Mereology'. *Stanford Encyclopedia of Philosophy*. online

Cotnoir, A. and A. Varzi (2021): *Mereology*, Oxford UP, Oxford.

Moltmann, F. (2016): '[Plural Reference and Reference to a Plurality. Linguistic Facts and Semantic Analyses](#)'. In M. Carrara / A. Arapinis / F. Moltmann (eds.): *Unity and Plurality. Logic, Philosophy, and Semantics*. Oxford University Press, Oxford, 2016, pp. 93-120.

Session 3:

The notion of an integrated whole and mereotopology, and the mass-count distinction

Reading:

Grimm, S. (2018): 'Grammatical Number and the Scale of Individuation'. *Language*

F. Moltmann (1998): '[Part Structures, Integrity and the Mass-Count Distinction](#)'. *Synthese* 116, 1998, 75-111

F. Moltmann: On the ontology and semantics of the mass-count distinction
'On the Semantics and Ontology of the Mass-Count Distinction'. In *Philosophy Compass* 20(3), March 2025; [link](#),

Session 4:

The view of the priority of the whole 1

Reading:

Carrara, M. And G. Landau (2025): 'Mereomodal Partialhood and Fractional Counting'.
Philosophical Studies, online first

Bennet, K. (2013): 'Having a part twice over'. *Australasian Journal of Philosophy* 91.1,
83 - 103

Liebesman, D. (2025): 'Partialhood'. In K. Bennett and D. Zimmermann (eds.): *Oxford Studies in Metaphysics*. Vol. 14, 131-175, Oxford University Press, Oxford.

<http://doi.org/10.30687/Jolma/2723-9640/2024/02/003>

J. Schaeffer (2010): 'Monism: The Priority of the Whole'. *Philosophical Review* 119.1.,
31-75.

Session 5:

The view of the priority of the whole 2

Reading:

F. Moltmann 'On the Ontology and Semantics of Absence', *The Journal for the Philosophy of Language, Mind, and the Arts* JOLMA 5.2., 2024

P. Ardisson, F. Dikmen, K. Mueller, F. Moltmann (2025): '[The Role of Plans for Qua Events and their Effects on Part Structure](#)'. *Proceedings of Formal Ontology in Information Systems* (FOIS) 15, IOS Press, Amsterdam.

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Handout 1

Ontological and Semantic Issues concerning Part-Whole Structure

1. Evidence for part-whole structure

Unproblematic facts

Entities generally have parts.

Entities may come with a structure organizing the parts and thus form a unified whole.

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But what sort of evidence do we have for what counts as a part and for what part-whole structures are? One way of approaching the topic:

Look closely about how we talk about parts and wholes.

Linguistic expression generally focused on *part of*

Part of denotes spatial inclusion?

Examples with *part of* from Varzi (*Mereology* SEP 2016):

- (1) a. The handle is part of the mug.
- b. The remote control is part of the stereo system.
- c. The left half is your part of the cake.
- d. The cutlery is part of the tableware.
- e. The contents of this bag is only part of what I bought.
- f. The outermost points are part of the perimeter.
- g. The first act was the best part of the play.

(1a): functional part (replaceable)

(1b) spatially discontinuous part

(1c): material part, or indeterminate part; compare

(1c): attention: there is also an intensional use of *half*:

- (1) c'. Half of the cake is yours.

c''. ?? Half of the cake is part of the cake. (any half)

(1d) part of a collection, spatially independent

(1e) *part of* applies unspecified type of material (or unspecific) entity

(1f) parts of abstract object

(1g) part of an event.

2. Standard, but not unproblematic, conditions on parthood

(2) a. Everything is part of itself. (reflexivity)

b. Pxx

(3) a. Any part of any part of a thing is itself part of that thing. (transitivity).

b. $(Pxy \wedge Pyz) \rightarrow Pxz$

(4) a. Two distinct things cannot be part of each other. (antisymmetry)

b. $(Pxy \wedge Pyx) \rightarrow x = y$

Some definitions

(5) Proper parthood : $PPxy =_{df} Pxy \wedge \neg x=y$

(6) Proper extension: $PExy =_{df} Pyx \wedge \neg x=y$.

(7) Overlap : $Oxy =_{df} \exists z(Pzx \wedge Pzy)$

(8) Disjointness : $Dxy =_{df} \neg Oxy$

Some issues with the standard conditions:

Reflexivity

Counterintuitive? Not so for the expression *part of*:

(9) a. John owns part of the house, maybe even all of it.

b. John read part of the book, and in fact all of the book.

Quantification over parts:

(10) All of the students were praised. $\rightarrow$ The students as a whole as well?

Transitivity

Well-known problem cases

Examples from Moltmann (1997)

Invalid inferences:

(11) John is part of the class

John's leg is part of John.

John's leg is part of the class.

(12) The page is part of the book

The book is part of the library.

The page is part of the library.

Does the integrity of the intermediary item block transitivity?

No: it depends on the nature of the whole:

(13) The (nonempty) page is part of the book.

The book is part of Kant's written work.

The page is part of Kant's written work.

But no transitivity with 'the empty pages of the book', 'the margins of the page', etc.

Another example with individuals and functional parts

(14) The arm is part of the body.

The hand is part of the arm.

The hand is part of the body.

Other cases of failure of transitivity:

Furniture: not all parts of a piece of furniture are furniture again.

Luggage: a handle of a suitcase is not luggage.

Ways of dismissing problems for transitivity

1) do not count functional parts as the only parts

Issue: what is the intuitive basis for the part relation, if not the applicability of *part of*?

2) distinguish different part relations, for different ontological levels:

Part relation for individuals, part relation for pluralities, part relation for stuff

Issue: transitivity problems arise within the level of individuals as well (14).

A different perspective: the priority of the whole

Functional and form-related parts depend on the nature of the whole.

For certain entities what the parts are depends on the nature of the whole, its structure, functional organization etc.

Extensionality

Additional principle:

(15) Strong supplementation

$$\neg Pyx \rightarrow \exists z(Pzy \wedge \neg Ozx)$$

From (2-4) and (15) we can derive Extensionality:

(16) Extensionality:

Composite objects with the same proper parts are identical:

$$(\exists zPPzx \vee \exists zPPzy) \rightarrow (x = y \leftrightarrow \forall z(PPzx \leftrightarrow PPzy)).$$

Unproblematic:

$$(17) x = y \rightarrow \forall z(PPzx \leftrightarrow PPzy)$$

(17) is an instance of the indiscernability of identicals:

$$(18) x = y \rightarrow (\phi x \leftrightarrow \phi y)$$

More problematic:

$$(19) \forall z(PPzx \leftrightarrow PPzy) \rightarrow x = y$$

Problem cases for extensionality

Diachronic example: chair rebuilt as a table

Synchronic examples:

two committees with the same members, but different functions, musical quartet and a sports team, collection of dots, heart drawn with dots, piece of furniture and work of art,

Constitution

A sculpture and the wood that constitutes it: have different temporal and modal profiles and differ in other properties.

Diagnosis of the issues

Extensionality should not hold if the whole is also individuated by structure and (practical, aesthetic) function.

Atomicity

$$(20) \text{ a. Atom: } Ax =_{\text{df}} \neg \exists y PPyx$$

$$\text{ b. Atomicity: } \exists y (Ay \wedge Pyx)$$

$$\text{ c. No being an atom: } \exists y PPyx$$

The notion of an atom in natural language semantics

Individuals with respect to the plural-specific part-relation (discussion tomorrow!)

3. Different uses of *part* / different nouns *part*

The mass-count distinction for *part*

Part of vs. *a part of* (e.g., Moltmann 1998)

- (21) a. John and Mary are part of the class.
b. ?? John and Mary are a part of the class.
- (22) a. Sugar is part of lemonade
b. ??? Sugar is a part of the lemonade.
- (23) a. Joe ate part of the apple.
b. ?? Joe ate a part of the apple.

The light noun *part* (Ardisson et al. 2025)

Full noun *part*

- (24) a. Parts of the ceremony took place yesterday.
b. What John did was a part of the ceremony.
- (25) a. John and Mary took part in the ceremony.
b. The priest was part of the ceremony.
c. ??? The priest was a part of the ceremony.

Predicates of participation and the occurrence of the light noun *part*:

Take part in, part-icipate, German teil-nehmen, French faire partie, Italian avere parte

Is the occurrence of *part* accidental?

If not, what does this mean for the part structure of events?

4. What is the basis for deciding about the ontology of part-whole structure

1) Intuitions about entities and their part-whole structure

What sorts of evidence do we really have about our intuitions regarding reality?

Intuitions need to be uncovered; they are not generally straightforward.

One way of uncovering them is through linguistic data: acceptability judgments.

2) What do the linguistic data consist in?

Standard practice in natural language semantics appeals to:

1. Uses of *part of*
2. More indirect apparent linguistic involvement of part-whole structure: semantics of plurals and mass nouns, part-whole structure of situations used in situation-based semantics (truthmaker semantics)

1) and 2) are not generally taken to coincide

A common view among formal semanticists: compositional semantics requires a particular notion of part, which need not coincide with the notion of conveyed by explicit linguistic expressions such as *part*. Semantics of plurals and mass nouns requires extensional mereology, even if that is not reflected in uses of *part*.

A different approach

Take into consideration the whole range of part-whole-related expression in natural language
Different nouns *part, whole, individual, complete, partial, is missing, ...*

5. Three views of part-whole structure

Extensional mereology (e.g., Link 1983)

Part-whole structure consists in a part relation that is transitive, extensional and closed under sum formation

Mereotopology and hylomorphism (e.g., Moltmann 1997)

Part-whole structure also consists in conditions of form, structure, connectedness, boundedness

The priority of the whole (new – and very old)

The whole is prior to its parts: it depends on the whole what its parts are and how the parts are to be understood, in terms of function or form.

The views may apply to different sorts of objects and to the semantics of different expressions in natural language.

6. Part-whole structure in linguistics

6.1. Explicit expressions of part-whole-related expressions

Part of, a part of

Other partitive constructions: *all of, most of, none of, much/many of*

Do all partitive quantifiers pick out the same sorts of parts as *part of*?

Completion-related expressions

Partial(ly), complete(ly)

Nominal modifiers

- (23) a. partial copy of the paper
b. a complete copy of the paper
- (24) a. partial realization of the plan
b. complete realization of the plan
- (25) a. partial reconstruction of the church
b. complete reconstruction of the church
- (26) partial truth, complete truth

Adverbial use

- (27) a. John partly forgot the poem.
b. John completely forgot the poem.

Issue

Reference to concrete object (original)

Reference to abstract object (event)

Completion of a whole and absence of a part

Complete vs. *is missing* NP

- (28) a. The collection is complete.
b. The collection is missing a piece.
- (29) a. The quotation is complete.
b. The quotation is missing a few words.
- (30) a. The house is complete.
b. The house is only missing the windows.

Other expressions of completeness

Whole:

- (31) a. The whole collection is expensive. (every part of the collection as a whole)
 b. The whole thing is incomprehensible. (every part and overall)
 c. the whole plan was misguided (every part and overall)
 d. the whole time (every part)

German has two words for ‘complete’: *voellig* and *vollstaendig*

- (32) a. die vollstaendige Uebersetzung
 ‘the complete translation’
 b. ??? die voellige Uebersetzung
- (33) a. die voellige Dunkelheit
 ‘the complete darkness’
 b. ??? die vollstaendige Dunkelheit

Vollstaendig involves a reference object with discrete parts, but not so *voellig*, which relates to a homogenous part structure.

Conclusion

Natural language displays a range of expressions involving not just a part-of-relation, but also reference to concrete or abstract wholes and the structure of a whole.

6.2. Plurals and mass nouns

Parallels between singular count, plural, and mass NPs.

Predicates

Singular count, plural, and mass NPs can be arguments of the same predicates.

Determiners

- (34) a. the / some woman (singular count)
 b. the / some women (plural)
 c. the / some personnel (mass)

Conjunction

- (35) a. The man and the women met.
 b. The men and the women met
 c. The sugar and the cinnamon were mixed.

Partitive construction

- (36) a. part / some / all / most of the apple
 b. part / some / all / most of the apples
 c. part / some / all / most of the sugar.

Connections to event semantics

Parallel effect of singular count, plural and mass NPs on the applicability/understanding of *in*-adverbials and *for*-adverbials:

- (37) a. John ate the apple in two minutes.
 b. John ate the peas in two minutes.
 c. John ate the soup in two minutes
 (38) a. ??? John ate the apple for an hour
 b. ??? John ate the apples for an hour.
 c. ??? John ate the sugar for an hour.

Standard view about the semantics of plural and mass nouns

Extensional mereology

Three domains with their own part relations:

1. The domain of individuals with a part relation among individuals
2. The domain of pluralities, the set of sums of individuals, with a part relation that is closed under sum formation

Individuals as atoms with respect to the plural-specific part relation

3. The domain of quantities with its part relation that is closed under sum formation, no atoms

Definite plurals refer to sums of individuals

- (39) [*the children*] = sum([*child*])

Definite mass NPs refer to sums of quantities

- (40) [*the water in the glass*] = sum ([*water in the glass*])

Two diverging semantic requirements

- 1) Analogy between singular count, plural, mass semantics
- 2) Counterintuitive consequences of a singularist semantics of plural and mass NPs:

Pluralities treated as single entities.

Quantities treated as single entities.

But:

- (41) a. John and Mary are two.
 b. The children are ten.
- (42) a. ??? The men and the women are two.
 b. ??? The water and the wine are two.

Generalization

Pluralities and quantities never count as single entities, but the formal semantics treats them as such.

Remedy in the case of plurals

Plural reference: definite plural NPs refer to several individuals at once, rather than referring to a single collective thing (Oliver and Smiley 2013, Yi 2005/6, McKay 2006, ...)

Remedy in the case of mass NPs:

Sui generis mass reference? (McKay 2016)

Issue

How is plural reference (sui generis mass reference) compatible with the semantic analogy of singular count, plural and mass NPs?

Some references

Cotnoir, A. and A. Varzi (2021): *Mereology*. Oxford UP, New York.

Varzi, A. (2016): 'Mereology', *The Stanford Encyclopedia of Philosophy*, Edward N.

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Moltmann, F. (1997): *Parts and Wholes in Semantics*, Oxford UP, New York.

[Online version](#)

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Handout 2

The Extensional Mereological Approach to Part-Whole Structure

Recapitulation of last time

Ontology of part-whole structure:

What is the basis for deciding about ontology and about part-whole relations in particular?

Linguistically manifest intuitions about part-whole structure; but what sorts of linguistic data are relevant?

Varzi's data: sentences with *part of*

Other explicit part-whole related expressions:

Partial, complete: reference to an abstract or conceived whole

Different kinds of complete wholes: German *voellig* vs. *vollstaendig*

Indirect function of part-whole structure:

Semantics of definite plural and mass NPs, conjunctions of definite NPs, connections between the semantics of NPs and event semantic connections

Overall conclusion

Linguistic data about part-whole structure are much richer than generally considered by mereologists.

1. Extensional mereology

Extensional mereology, informally

- Part relation is transitive.
- Part relation is closed under sum formation (fusion): unrestricted sum formation.
- Entities that share the same proper parts are identical.

Suited for completely unstructured wholes

Formal conditions of extensional mereology

- (1) a. Reflexivity: $\forall x(x < x)$
- b. Transitivity : $\forall x \forall y \forall z((x < y \ \& \ y < z) \rightarrow x < z)$
- c. Antisymmetry: $\forall x \forall y((x < y \ \& \ y < x) \rightarrow x = y)$
- d. Extensionality : $\forall x \ z \forall x \forall y((x < z \ \& \ x \neq z \ \& \ y < z \ \& \ y \neq z) \rightarrow x = y)$

Sum formation or fusion *sum*

Definition based on overlap:

- (2) Definition of overlap
- $x \text{ O } y =_{\text{def}} \exists z(z < x \ \& \ z < y)$

- (3) Definition of sum formation

For a nonempty set P, $\text{sum}(x, P) =_{\text{def}} \forall y(y \text{ O } x \leftrightarrow \exists z(y \text{ O } z \ \& \ P(z)))$

I.e., a sum of a set P is a thing such that everything which overlaps with it also overlaps with something in P, and vice versa.

Here ‘sum’ denotes a relation between an entity and a sum,

Alternatively: sum as an operator applying to sets, uniqueness guaranteed by Uniqueness of Sums:

- (4) The axiom of uniqueness of Sums (= Extensionality)
- Two things composed of the same parts are identical.
- $\forall P(P \neq \emptyset \rightarrow \exists! z \text{ sum}(z, P))$

Other notations:

- (5) a. Binary sum: $x \oplus y =_{\text{def}} \exists z \text{ sum}(z, \{x, y\})$
 - b. Generalized sum: for any nonempty set P, $\oplus P =_{\text{def}} \iota z \text{ sum}(z, P)$.
-

2. Application of extensional mereology

Unified semantics of definite NPs using sum formation

- (6) a. the chairman
- b. $\iota x(\text{sum}(x, C)) (= \iota x C(x))$

(7) a. the students

b. $\iota x \text{sum}(x, S)$

(8) a. the water

b. $\iota x \text{sum}(x, W)$

The ι -operator 'the'

$\iota x P(x)$ is defined only if P holds of exactly one individual, and when defined, it denotes that individual.

Non-Boolean conjunction: conjunction interpreted by sum formation

Boolean conjunction: &

(9) a. John and Mary embraced.

b. The men and the women gathered.

c. The oil and the vinegar were mixed.

(10) a. $j \oplus m = \iota x \text{sum}(x, \{j, m\})$

b. $(\iota x \text{sum}(x, M)) \oplus (\iota x \text{sum}(x, W)) = (\oplus M) \oplus (\oplus W) = \iota x \text{sum}(x, M \cup W)$

c. $(\iota x \text{sum}(x, O)) \oplus \iota x \text{sum}(x, V) = (\oplus O) \oplus (\oplus V) = \iota x \text{sum}(x, O \cup V)$

(11) a. $E(j \oplus m)$ (formalization of 10a)

b. $G(\iota x \text{sum}(x, M \cup W))$ (formalization of 10b)

c. $M(\iota x \text{sum}(x, O \cup V))$ (formalization of 10c)

Some issues with sum formation

Sum formation does not apply to parts of individuals to yield individuals:

(12) a. the hot dog = the sausage and the bun?

b. The sausage and bun taste similar

c. The sausage and the bun cost 4 dollar.

(13) a. The hot dog tastes similar.

b. The sausage and the bun costs 4 dollar.

But compare adjective conjunctions:

With adjective conjunctions, sum formation is applicable to single individuals

(14) The Italian flag is red, green and white.

Adjective conjunction seems to be domain-neutral:

- (15) a. the blue and white *pillow*
 b. the blue and white *pillows*
 c. the blue and white *laundry*

Other sorts of predicate conjunctions vs. NP-conjunctions

States:

- (16) a. John's being calm and nervous $\neq$ John's being calm and John's being nervous
 b. John's being calm and John's being nervous do not go together.
 c. ??? John's being calm and nervous does not go together
- (17) a. John's hope to win and to become rich makes sense.
 b. John's hope to win and his hope to become rich make sense.

Events:

- (18) a. John's walking up and down showed his impatience / lasted for an hour.
 b. John's walking up and his walking down showed his impatience / lasted for an hour.

Predicate conjunction can be used to characterize complex states or events.

NP-conjunction can only lead to pluralities of states or events.

The state of being happy and calm vs the states of being happy and being calm

The act of walking up and down vs. *the act of walking up and the act of walking down*.

Conclusion

Non-Boolean NP-conjunction interpreted by sum formation can only lead to entities in the domains of pluralities and quantities (the mass domain), not individuals (denotable by singular count NPs).

Non-Boolean predicate conjunction is interpreted in a domain-neutral way.

3. More on extensional mereology

The lattice theoretic (algebraic) perspective (Link 1983)

The domain of pluralities and the domain of quantities form join semi-lattices

Join semilattice $(L, \oplus)$

For any two elements $x, y \in L$, the join $x \oplus y$ is defined, and $x \oplus y \in L$.

Join operation of a lattice meets the following conditions

$$(19) \text{ a. associativity } (x \oplus (y \oplus z)) = (x \oplus y) \oplus z$$

$$\text{b. Commutativity } (x \oplus y) = (y \oplus x)$$

$$\text{c. Idempotence } (x \oplus x = x)$$

Impose an additional condition :

$$(19) \text{ d. Unique Separation: } x < y \rightarrow \exists! z (x \oplus z = y \ \& \ \neg(x \circ z))$$

$<$ and $\oplus$ are interdefinable:

$$(20) \ x < y \iff x \oplus y = y$$

With Unique Separation, the join operation $\oplus$ will have the same properties as the sum operation $\text{sum}_<$ in mereology.

Complete join semilattice:

A join semilattice where every subset has a join (with bottom element (join of empty set) being removed)

4. The formal semantics of plurals and mass nouns

Extensional mereology requires distinguishing three domains of entities with three distinct part relations:

The domain of individuals I , ordered by $<$ (part of individuals)

The domain of pluralities P , ordered by $<_i$ (parts of pluralities)

The domain of quantities (or portions or stuff) M , ordered by $<_m$ (parts of quantities)

$(P, \oplus_i)$ is a join semilattice i.e., $(P, <_i)$ is an extensional mereology

$(M, \oplus_m)$ is a join semilattice, i.e., $(M, <_m)$ is an extensional mereology

Important:

Necessity to distinguish two part relations $<_i$ and $<_m$, in addition to the part relation $<$ applying to parts of individuals.

Example:

A part of something described as a ‘sum’ is neither a part in the sense of $<_i$ nor in the sense of $<_m$, but only in the sense of $<$.

A part of a plurality is never a part in the sense of $<$ or $<_m$.

Individuals are atoms with respect to $<_i$, but not with respect to $<_m$, and $<_m$ does not apply to pluralities or individuals (except in Link 1983).

The semantics of singular count, plural, mass nouns

- (21) For a singular count noun extension N ,
for all x , $N(x)$, x is an atom with respect to $<_i$.
- (22) $\text{atom}(x) =_{\text{def}} \neg \exists y (y < x \ \& \ x \neq y)$

Potential problems: sequence-type nouns

Continuous parts of sequences are again sequences

Similarly for sums, entities, quantities, fences, walls, Russian dolls

A problem considered a major challenge to the notion of an individual (and for the extensional mereological account of the mass-count distinction): see S. Rothstein (2017): *Semantics of Counting and Measuring*.

However, sequence-type nouns do not actually pose a problem:

A part of a sequence is not a part in the sense of $<_i$, but only in the sense of $<$, since a sequence is not a proper plurality.

A sequence and its parts are atoms – with respect to $<_i$!

(23) The semantics of plurals within extensional mereology

For a plural noun extension N_{pl} , $N_{pl} = \oplus N$

Exclusive and inclusive conceptions of pluralities

Do pluralities include pluralities of one? Good evidence that they do.

- (24) Does John have children? Yes, he has one daughter.

Plural extensions are cumulative and atomic

- (25) $(N_{pl}, \oplus_i)$ is a complete join semilattice

(26) The semantics of mass nouns in extensional mereology

For mass noun extension N , $(N, \oplus_m)$ is a complete join semilattice.

Mass nouns extensions are cumulative, but not atomic.

Are mass noun extensions divisive?

(27) A set N is divisive $\stackrel{\text{def}}{=} \forall x \exists y (y < x \ \& \ y \neq x)$

Problems

Minimal parts problem: proper parts of H_2O molecules are no longer water.

Object mass nouns:

furniture, luggage, glassware, clothing, faculty, police force

Minimal pairs:

clothes – clothing, policemen – police force, cows – cattle, pâtes – pasta

Mass noun extensions are merely perceived as being divisive?

Mass nouns extensions are merely realizations of a type that is specified as homogenous, without having to be homogenous themselves (Liebesmann ‘Partialhood’, 2024).

General difficulties for extensional mereology

Can the notion of an atom account really for countability, for being a single object?

In extensional mereology being an atom means: being denoted by a singular count.

Extensional mereology with its distinction into plural-specific and mass-specific domains involves strictly language-dependent part structure, dependent on the use of plural or mass categories.

This means it does not really provide a content of the morphosyntactic mass-count distinction or it does not appeal to a language-independent notion of being ‘one’, a single object.

5. Arguments for a single part relation (Moltmann 1997, 1998)

1. Uniform semantics of adjective conjunction

(28) Whatever is in the bag is white and red.

2. Special quantifiers across domains

Special quantifiers: *something, everything, what, that* (see e.g., Moltmann 2022)

Special quantifiers are domain-neutral:

- (29) a. *What* is in the bag? An apple, some bread, some coins
 b. John ate *something*, bread, an apple, or some cookies.
 c. John ate something the bread, the apples, the ten cookies.

Application of part-related expressions to special quantifiers:

- (30) John ate part of what was in the bag (part of an apple, part of some bread, part of some cookies)

3. Modifiers with domain-independent part structure-related content

Part structure-sensitive adjectives

- (31) a. the frequent rain
 b. the frequent rainfalls

Meaning of *frequent* involves temporal separation of parts, in the generalized sense of ‘part’.

4. Domain-independent part quantifiers across domains: *whole* across languages

French:

- (32) a. toute la journée ‘the whole day’
 b. tous les jours ‘the whole days’
 c. tout le vin ‘the whole wine’

German:

- (33) a. der ganze Tag ‘the whole day’
 b. die ganzen Tage ‘the whole days’
 c. der ganze Wein ‘the whole wine’

5. The noun *part* as a domain-neutral expression:

- (34) a. John ate part of the apple / the fruit / the grapes
 b. John ate the apple / the fruit / the grapes only in part.
-

6. Extensional mereology and event semantics

Verbs have cumulative extensions.

Different thematic relations require cumulativity:

(35) a. John laughed.

b. The children laughed

(36) a. John drank the wine in the first glass.

b. John drank the wine in the first and the second glass.

Measuring out events:

(37) a. John drank the wine in the two classes in ten minutes.

b. or two hours John drank wine.

Events may inherit part structure from event participants.

Multidimensional part structures may result:

(38) The guests drank the two bottles of wine.

However, the verbal domain of events does not seem to divide into singular count-plural-mass semantically.

Rather it acts like a classifier system - roughly like nouns in Chinese.

Verbs classify as mass with respect to the Davidsonian event argument position.

E. g., numerals require the classifier *times* when applying to any verb or VP (Moltmann 1997, to appear):

(39) a. * John fell three.

b. John fell three times.

(40) John worked out three times today.

Some suggested readings

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Handout 3

The Notion of an Integrated Whole and Mereotopology

1. The notion of integrated whole

1.1. The traditional notion of an integrated whole

The notion of an integrated whole - intuitive description:

Having a form, structure, boundary, being maximally self-connected

Being unified, being a single thing by being unified

The importance of the notion of form in ontology

Two entities are identical not just if they are constituted by the same matter, but also share the same essential form.

Two forms of identification

Identity throughout times (endurance)

Identity across different situations or worlds

Historically

Aristotle's hylomorphism: entities consist in parts and a form

Contemporary hylomorphism: Johnston (2006), Fine (2010), Koslicki (2008)

A major philosophical puzzle

What constitutes the unity of matter and form?

What makes something constituted by matter and a form a single thing?

Different views of the ontological status of form

Form as another part of an object: Koslicki 2008

Form as the unifying force combining the parts: Johnston 2006

Form as structure-building operations for wholes: Fine 2010

The notion of an integrated whole in natural language semantics

Cognitive semantics

Langacker (1987): notion of boundedness, having a boundary

Formal semantics

Moltmann (1997, 1998, 2005): introduction of the general notion of an integrated whole to natural language semantics and of the notion of a maximally self-connected whole in particular (mereotopology)

Mereotopology

studies integrated wholes in space or time, in particular the notion of an entity that is maximally self-connected in space or in time (R-integrated whole, for R being spatial or temporal contiguity)

Recent uses of mereotopology by other linguists (Grimm, Filip, Sutton, Henderson, ...).

The notion of an R-integrated whole (maximal self-connectedness)

R_{trans} : the transitive closure of R.

(1) Definition

For a nonlogical, symmetric relation R, x is an *(R-)integrated whole* ((R)-INT-WH(x)) iff

for every y and z such that $y < x$ and $z < x$, $R_{trans}(y, z)$, and for every y such that $y < x$, for no w, $\neg w < x$, $R_{trans}(y, w)$.

Special case

FF-integrated whole: an R-integrated whole for the relation FF defined as follows

(2) For a property F, $FF(x, y)$ iff $F(x)$ and $F(y)$.

Examples

(4) a. the blue things

F = being blue.

b. The children in the garden

F = child in the garden

Some linguistic applications

The readings of part-related predicates *compare* and *distinguish* (Moltmann 1997)

FF-integrated wholes:

- (5) a. John cannot distinguish the boys and the girls
- b. Mary compared the blue and green balls.

R-integrated wholes:

- (6) The students that share a dorm room will get the same key

R: share a dorm room with

- (7) a. Mary compared the furniture.
- b. Mary compared the furniture in the two rooms.

R: be in the same room as

The meaning of *times*

- (8) a. John fell several times today.
- b. John slept several times today.
- c. John was awake a number of times (when I looked into his room, which I did throughout the night).

8a: counting events that are inherent integrated wholes

8b: counting maximally continuous processes in time

8c: counting states at given occasions.

General questions

Are integrated wholes always objects?

Perhaps not:

Pluralities and subpluralities can be integrated wholes.

Quantities and subquantities can be integrated wholes.

Issues

Is the notion of an integrated whole restricted to objects?

Does it always pertain to essential properties of objects?

Essential properties

1. properties an object needs to have in order to exist
2. properties an object needs to have in order to be what it is

1.2. Three types of integrated wholes

1. Essential integrated wholes

Examples:

A heap of sand, a house, a tree, a statue, a ship, a person

Essential integrated wholes may allow for the replacement of parts or constituting material:

Famous example: the ship of Theseus

But not all essential integrated wholes allow for the replacement of parts:

A word as an abstract object

the couple of John and Mary

the group of the students

Kit Fine's (1999) notions:

Essential integrated wholes that allow for replacement of parts: *variable embodiments*

Essential integrated wholes that do not allow for replacement of parts: *rigid embodiments*

2. Accidental integrated wholes

The sand when taking the form of a heap

The clay when taking the form of a statue

The wood when having the form of a tree

3. Conceived integrated wholes

An object may be merely conceived or perceived as an integrated whole:

A notion used in cognitive semantics (Langacker 1987, Jackendoff)

Two uses of conceived integrated wholes in the literature:

Boundedness with a subjective center (Langacker 1987)

Merely conceived integrated whole in Moltmann (1998):

(9) a. the amount of sand (as opposed to 'the sand')

b. the collection of things over there

(10) a. the students together

b. the students as a whole

The claims:

Accidental and conceived integrated wholes are linguistically important:

Langacker (1987) (and Moltmann 1998):

Subjective boundedness plays a role for the mass-count distinction.

Another application in Moltmann (1997, 1998):

Modifiers such as *as a whole*, *as a team*, and *together* specify an entity as a merely conceived integrated whole, influencing the availability of distributivity and part-related predicates.

Pluralities and quantities as accidental integrated wholes influence the application of part-related expressions

(11) a. John gave the class as a whole an A.

b. John and Mary together / as a team will be able to solve the problem.

2. The importance of the notion of integrated whole for formal part-whole relations

2.1. Extensional mereology

Posits transitivity, unrestricted sum formation, unique sums (extensionality) for the part-relation

Consequence:

Needs to distinguish different part relations for individuals, pluralities and quantities

Mass-count distinction defined in terms of atomicity and cumulativity

Individual: an atom with respect to the plural-specific part relation:

Quantity: not an atom or not necessarily an atom with respect to the quantity-specific part relation

Domains of pluralities and quantities are cumulative, but not the domain of individuals.

Problems for the division into different part structures

Conceptual problem

Circularity:

Formulating the content of the mass-count distinction involves essential reference to domain-specific part relations, but those part relations strictly depend on the use of mass nouns, singular count nouns, plural nouns themselves

→ No language-independent way of conceiving of the content of the mass-count distinction available within extensional mereology.

Empirical problems

- (12) a. John ate part of/all of/some of what was on the table (the apple/the nuts/the bread).
- b. John did not eat part of what was on the table.
- (13) a. part of the apples and the milk
- b. part of the apple and the milk
- c. John ate part of what was on the table and what was in the fridge.

Define a new generic part relation $<_{\text{gen}}$ for mixed domains on the basis of individual-, plurality- and quantity specific part relation?

But this gives rise to problems of transitivity:

- (14) a. John painted part of the paintings.
 - b. John painted part of one of the paintings.
- (14b) does not entail (14a).

2.2. Conditions on part structures and integrated wholes

Transitivity:

- (15) The page is part of the book.
 The book is art of the library.
 The page part of the library.
- (16) The page is part of Goethe's written work.
 The book is part of Goethe's written work
 The page part of Goethe's written work.

Moltmann (1997, 1998):

- (17) Condition on transitivity in the domains of individuals and quantities
 (nonessential integrated wholes only):

If for pluralities, x, y, z , if $x < y$, $y < z$, then $x < z$ only if y is not an integrated whole.

Condition on the integrity of the intermediary item does not suffice.

For essential integrated wholes, it depends on the whole what its parts are (and thus whether the part relation is transitive with respect to the parts pertaining to that whole)

Addendum to Moltmann (1998):

For quantities it depends on what quantities are made of whether the part relation is transitive (*water, laundry, furniture, stuff*).

Integrates wholes and sum formation

(18) Restriction on sum formation

For a non-empty set X , $\text{sum}(X)$ exists only if $\text{sum}(X)$ would be an integrated whole.

Important:

The integrated whole permitting sum formation may be a merely accidental or conceived integrated whole:

- (19) a. The sum that is 'the children': accidental integrated whole
- b. A sum that is 'five children together': conceived integrated whole.

3. The mass-count distinction based on the notion of an integrated whole

3.1. Langacker (1987)

Singular count nouns describe bounded entity, but not so mass nouns

Boundedness in only one dimension: lines, fences

Boundedness with subjective center: flash of light, collection of stuff over there

Mass-count distinction

Singular count nouns denote bounded 'regions' (entities)

Mass nouns denote unbounded regions (entities).

Boundedness may pertain only in one dimension:

‘New’ solution to the problem of sequence-type nouns: sequences are bounded in one dimension, hence *sequence* is count.

3.2. Moltmann (1998)

- (20) a. For a singular count noun *N* and a reference situation *s*, if $N(x, s)$, then *x* is an integrated whole in *s*.
 b. For a mass noun *N* and a reference situation *s*, if $N(x, s)$, then *x* is not an integrated whole in *x*.

Important: there are different ways to be an integrated whole

An entity *x* is an integrated whole in a reference situation *s* if 1, 2, 3, or 4:

1. *x* is an *essential* integrated whole (in *s*)
2. *x* is an *accidental* integrated whole in *s*: *x* has properties defining it as an integrated whole perhaps only in the temporally restricted situation *s* or the circumstances of *s*.
3. *x* is *conceived* as an integrated whole relative to *s* (*s* being dependent on the mind of the speaker).
4. *x* is an integrated whole due to the particular *partial information* contained in *s*.

Accidental integrated wholes

- (21) ??? The line of people/The (loose) collection of papers on my desk/The group of people in the room does not exist anymore.

Conceived integrated wholes

- (22) a. This patch of snow is a new patch of snow.
 b. ?? This patch of snow is new snow.
 (23) a. This snow is new snow.
 b. ?? This snow is a new patch of snow.

Integrated wholes due to partial information

- (24) a. the blue things and the white things
b. the big things and the small things

Semantic mass-count agreement conditions:

Implicit integrity conditions with *thing* (Moltmann 1998)

- (25) a. This thing is an apple. (pointing at an apple)
b. ??? This thing is apple. (pointing at an apple or a piece of apple)
c. This stuff / The content of the bowl is apple. (pointing at small pieces of apples)

Implicit integrity conditions with *part*

- (26) a. This wood is part/ ??? a part of the chair. (pointing at a piece or at pieces of wood)
b. The leg is a part of the chair.
c. The leg and the back are parts of / ??? a part of the chair.

3.3. Integrated wholes and semantic selection

The linguistic relevance of accidental integrated wholes:

- (27) a. John compared the people / ?? the line of people.
b. John cannot distinguish the papers on my desk / ?? the loose collection of papers on my desk.
c. John counted the people in the room / ?? the group of people in the room.

(28) The Accessibility Requirement

A predicate or reading of a predicate that makes reference to the parts of an argument, but not the whole, can apply to an object x in a reference situation s only if x is not an integrated whole in s.

- (29) a. John restructured the schedule.
b. John reorganized the team.

- (30) a. Hans zahlte die Klasse durch.
‘John counted the class through’
b. Hans zaehlte ?? die Klasse / ?? die Studenten.

‘John counted ?? the class / ok the students.’

The semantic effect of part-structure modifiers

- (31) a. John and Mary as a team can solve the problem.
 b. The stamps as a collection cost 100 dollar.
 c. The boxes together are too heavy.

The function of *whole*

- (31) a. The whole collection is expensive.
 b. The collection as a whole is expensive.

Two readings of (31a), only one reading of (31b).

What does *whole* in (31a) do?

1. maps an integrated whole and a reference situation *s* to a mere collection of parts in a minimally different reference situation *s*’.
2. maps an integrated whole and a reference situation *s* to the collection of the parts *and* the form in a minimally different reference situation *s*’. (Moltmann 2005)

A different perspective: plural reference

The accessibility requirement as a requirement on a plural argument position.

Predicates making reference to the parts, but not the whole require pluralities as many.

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Handout 4

The Priority of the Whole 1: Ordinary Objects and Predicates of Completion

1. The priority of the whole in our ordinary ontology

1.1. Examples

Artifacts

Function and form of parts to be understood only

House – door, window, balcony, chimney

Social objects

University, department faculty, student body

University building: main building, lecture halls, mensa,

Mathematical shapes

Names for parts of mathematical objects

Semicircle – half a circle

Hypotenuse – longest side of a triangle

Fractional amounts

half a liter milk

half a loaf of bread

1.2. What is the notion of priority?

Observation

Parts can exist without wholes:

Balconies, windows, doors, handles, arms, bones etc.

Two notions of ontological dependence

Existence dependence:

x is ontologically dependent on y iff x exists only if y exists.

Identity dependence:

x is ontologically dependent on y

iff the nature of x can only be understood in terms of the nature of y

iff the real definition of y mentions / may mention x.

My view of priority of the whole view

Only some wholes are prior to the parts.

Priority means identity dependence.

Contrast to historical views of monism

Monism:

The only ontologically independent entity is the cosmos.

Spinoza:

Only god is ontologically independent; everything is a trope (particularized property) of god.

2. The priority of the whole with events: plans and planned events

2.1. Plans

2.1.1. Reference to plans

(1) The plan to build a house

Kinds of event plans

Projects, intentions, future-directed dreams, instructions, recipes

Plans as enduring objects

Different existence predicates for different entities

- (2) a. The house still exists.
b. ??? The rain still exists.
- (3) a. The rain is still going on.
b. ??? The house is still going on.
- (4) a. The accident happened / occurred yesterday.
b. ??? The house happened / occurred last year.
- (5) a. An elaborate plan now exists.
b. The plan still exists to build a new house.
c. The composition / design / exists.
- (6) ??? The plan occurred / never happened.

Have as expressing dependent existence (of plans)

- (7) a. She has a plan to build the house.
b. She no longer has the plan to leave.
c. * She has (her) building the house.

2.1.2. What are the parts of plans?

Plans as abstract objects

Types of acts naturally act as parts of plans

- (8) To do write an article is part of the plan.

How does a plan look?

Mentions actions, their types and their connections

Mentions participants, agents, etc.

Mentions times and places

Participants as parts?

Times, places as parts?

Plans and their realization

- (9) a. realize a plan / project / dream
b. follow an instruction

c. play / perform a composition

2.2. Planned events

Examples: demonstrations, lectures, meetings, dances, competitions, ceremonies, masses

Distinguishing properties of planned events

1. Planned events go only with the existence predicate take place, not happen or *occur*.

- (10) a. The ceremony / demonstration / meeting took place / ?? happened /
 ?? occurred.
- b. The accident happened / occurred / ?? took place.

2. Planned events may allow for predicates of completion.

- (11) a. John completed the ceremony / the project / the dance.
- b. ??? John completed the accident / the hour-long shouting.

3. Planned events allow for predicates of participation.

- (12) a. John took part / participated in the ceremony / the demonstration / the project.
- b. ??? John took part / participated in the accident / the sudden spontaneous
 shouting.

Comitatives:

- (13) a. John's project / dance / lecture with Mary
- b. ??? John's accident / making noise with Mary

Roles of participants in events

Slots (Bennett 2007)

Slots as parts of plans?

Plans have slots that are specified for participants, are open, or partially specified

2.3. The light noun *part* (PART)

Occurs in predicates of participation

- (14) a. John took part in the ceremony.
- b. The priest was part of the ceremony.

take part in German : *teil-nehmen* ‘part take’,
 in Italian *avere parte*, ‘have part’,
 in French *faire partie* ‘make part’.

PART in predicates of sharing

German *teilen* and French *partager*

Implicit lexical component of English *share* ?

- (15) a. Hans teilte das online Dokument mit uns.
- b. John shared the online document with us.

- (16) a. Maria teilte ihre Erfahrungen.
- b. Mary shared her experiences.

‘Ordinary’ uses of predicates of sharing:

- (17) a. John shared the cake.
- b. John shared the task / the workload among his coworkers.

German *teilhaben* ‘have part’ reserved to conveying the sharing of experiences:

- (18) a. Maria liess uns an ihren Erfahrungen teilhaben.
- ‘Mary let us have part in her experiences.
- b. ??? Maria liess uns an dem Kuchen teilhaben.
- ‘Mary let us have part in the cake’

The full noun *part*

including those with the ordinary, inflected noun *part*:

- (19) a. The parts of the ceremony all took place in the village center.
- b. The parts of her experience that shocked me most concerned the interrogation.
- (20) a. ??? We had parts in / of the ceremony.
- (meaning: we took part in the ceremony)
- b. ??? We had parts of her experiences.
- (meaning: we were let have part in her experiences)

Ontology

Planned events as compositions, in some sense, of basic events and plans

Fine's (1982, 2022) operation of qua object formation

For a plan p and a basic event e realizing p , the planned event = e qua $\lambda x[x \text{ realizes } p]$ (e/p)

- (21) a. For a plan p and an event e that realizes p ,
the planned event e qua $\lambda x[x \text{ realizes } p]$ (e/p) exists.
b. For plans p and p' realized by events e and e' respectively, $e/p = e'/p'$ iff $e = e'$
and $p = p'$.

The semantics of *part(s)* and PART

- (22) a. For a planned event e/p , $[PART](e/p) = \{s \mid s \text{ is a (slot) part of } p\}$
b. For a planned event e/p , $[parts](e/p) = \{e' \mid e' < e\}$ ($<$ is the ordinary relation)

Other cases:

- (23) a. Parts of the university are in the city center. (spatial parts)
b. ?? Parts of the university are its physics and its math department.
(organizational parts)
c. These buildings / # The physics and the math department are the main parts of
the university.
d. The physics and the math department are part of the university.

Dual part structures for planned events, social objects ?

3. Expressions of completion: *Complete* and *partial*

3.1. The verb *to complete*

- (24) John completed the project, the ceremony, the recitation.

German:

- (25) Hans hat da Project *zuende gefuehrt*.
'John completed the project / lead the project to its end.'
(26) Hans hat das Projekt *beended*.
'John has terminated the project'.

(27) a. Hans hat das Projekt *vervollstaendigt*.

‘He added suitable abstract parts (and realized them).’

b. Hans hat den Plan *vervollstaendigt*.

‘John added parts to the plan that should have been there’.

(28) a. Maria hat das Gedicht *vollendet*.

‘Mary finished the poem’ (masterful whole)

b. Das Gedicht blieb *unvollendet*.

‘The poem remained unfinished.’

3.2. The predicates *complete* and *partial*

Two kinds of completion

1. Partial / full match of a concrete whole that is an original (in the present or the past)

(29) a. partial copy of the paper

b. a complete copy of the paper

(30) a. partial reconstruction of the church

b. complete reconstruction of the church

2. Partial / full manifestation of a merely conceived whole

(31) a. partial realization of the plan

b. complete realization of the plan

(32) a. John partly forgot the poem.

b. John completely forgot the poem.

Complete and *be missing* display the same kinds of interpretations.

Relative to an original

(33) a. The quotation is complete.

b. The quotation is missing a few words.

Relative to an actual whole

(34) a. The collection is complete.

b. After the break-in, the collection is missing a piece.

Relative to a conceived whole (design)

(35) a. The house is complete.

- b. The house is only missing the windows.

(36) The meaning of *complete*

For a concrete or abstract whole X and an entity d , $\text{complete}(d, X)$ iff d is a full realization of X .

The meaning of *partial*

Very tricky!

- (37) a. a partial house
b. a part of a house
- (38) a. a partial portrait of Joe
b. a part of a portrait of Joe
- (39) a. a partial copy of the page
b. a part of the copy of the page.
- (40) a. a partial realization of the plan
b. a part of the realization of the plane.

(41) Meaning suggestion for *partial*

For a concrete or abstract whole X and an entity d , $\text{partial}(d, X)$ iff d is a realization of part of X and is sort of the same type as X .

In the literature:

Liebesman (2024); partialhood as primitive

Carrara and Lando (2025): partialhood = parthood relative to a possible entity

3.3. Application to partial truth in truthmaker semantics

- (42) A sentence S is partly true iff for an exact truthmaker s of S ,
there is an s' , $s' < s$ and s' is actual.

Intended application:

- (43) It is raining and it is cold.

Truthmaker of 'R & C': the fusion of two situations s' and s'' .

‘R & C’ is partly true in case for an exact truthmaker s of ‘R & C’, there is a (conjunctive) part s' of s that is actual.

Truthmaker of ‘R & C’: the fusion of two situations s' and s'' .

(44) John loves Mary:

John loves Mary would be partly true just because John exists!

But if ‘<’ in (42) is understood as partialhood, this may require s' to be of the same type as s , thus it needs to be a situation and cannot just be an object.

3.4. The adverbial *partly*

(45) a. John partly solved the problem.

b. $\exists e \exists E$ (solved(E , John, the problem) & partly(e , E))

Question:

Where does quantification over abstract events come from?

Same as for the progressive?

(46) John was solving the problem.

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Approaches to Parts and Wholes in Semantics

Advanced Course

ESSLLI 2025 Summer School

Friederike Moltmann

Handout 5

The Priority of the Whole 2: Expressions of Replacement

Expressions of replacements in English

Verbs of replacement *replace*, exchange

Instead-phrases: *instead of Mary*, the adverb *instead*

In place-phrases: *in your place*

1. The verb *replace*

The role of wholes for replacement:

Replace applies to an entity relative to a contextually given *structured whole*,

Kinds of structured wholes

Artefacts

- (1) a. John replaced the leg (of the table).

Structured spatial region

- (1) b. John replaced the picture on the wall.

Social institution

- (1) c. John replaced the chairman (of the company).

Planned event

- (1) d. John replaced the priest (for the ceremony).

Configurations of individuals with their alienable possessions:

- (1) e. John replaced his computer / his trainer / his glasses.

Replace targets structured wholes that come with a particular role or place – a *slot*
replacement: exchange the *replacee* in the slot with a *replacer*.

Constraints on structured wholes:

- (2) a. ???John replaced the flower he had bought.
- b. ???John replaced the person speaking to his wife.
- c. ???John replaced the person that was spontaneously addressing the crowd.

Replacement can only target entities occupying slot within the given whole.

Replace cannot target the temporal or spatial location of the entity itself:

- (2) d. ??? John replaced the location of the company / the time of the event / the ceremony.

Reason:

Entities have a location in time and space, but that location does not occupy a slot within the entity.

Replace can target variable that fails to have a realization:

- (3) a. John replaced the missing screw.
- b. Hans hat die fehlende Schraube ersetzt.

The verb *exchange*

- (4) a. John exchanged the flower he had bought
 - b. John exchanged the missing screw.
 - (5) a. John exchanged his pen.
 - b. John replaced his pen. (suggests: because it was defective)
 - (6) a. John exchanged the overqualified employee.
 - b. John replaced the overqualified employee
 - (7) a. They exchanged information / documents.
 - b. ??? They replaced information / documents.
-

2. *Instead*-phrases

Syntax of *instead*-phrases: take nominal complements (NPs, APs, gerunds)

Two constructions with *instead*-phrases:

2.1. Constituent-based *instead*-phrases

Instead-phrase relates to particular constituents in the sentences by *association with focus*:

- (8) a. Instead of *Mary*, Bill met *Sue*.

- *Instead*-phrase applies to a structured whole that is a *structured proposition* or *state of affairs* expressed by the sentence

- Conveys replacement of entities that would fill the slots in that proposition that syntactically correspond to the focused constituents;

Alternative condition: the replacee will have to come from the focus-semantic value of the replacer

With *instead*-phrases, the replacer need not be an object, but can be

- a higher-order semantic value, e.g., a quantifier:

(8) b. Instead of a *man* the firm hired a *woman*.

- a relation:

(8) c. Instead of *running*, John *walked* to the house.

d. Instead of *over* the house, the bird flew *into* the house.

- several entities, relating to distinct positions in the structured proposition at once, e.g., with gapping:

(8) d. Instead of giving *John a book*, Mary gave *Bill a pen*.

Unlike the verb replace, *instead*-phrases may convey replacement of spatial and temporal locations:

(9) a. Instead of *Monday*, they arrived *Tuesday*.

b. Instead of in the *garden*, they celebrated in the *house*.

Reason:

Temporal and spatial locations when conveyed by adverbials occupy places in a structured proposition just like individuals.

2.2. Situation-based instead-phrases

Instead-phrase relates to a situation that is entirely given by the context:

(10) a. Instead of raining, it snowed.

b. Instead people dancing in the street, there was silence everywhere.

c. Instead of John praising himself, Joe criticized Mary.

Background situation consists of circumstances in which an event is expected to occur but does not, and in which alternative event does occur

Thus, a situation containing a slot for an event.

Instead-sentences seem restricted to event descriptions:

(11) a. ??? Instead of knowing / speaking French, John knows / speaks Spanish.

- b. Instead of speaking French, John spoke Spanish during the meeting.

2. 3. The anaphoric adverb *instead*

Instead without complement: anaphoric *instead*

1. when associating with focus:

relates anaphorically to the replacee in a preceding negative sentence:

- (12) a. Bill did not meet *Mary*. *Instead*, he met *Sue*.
 b. The firm should have hired a *woman*. *Instead*, it hired a *man*.
 c. John did not *run* to the use. *Instead*, he *walked* to the house
 d. The bird did not fly *over* the house. *Instead*, it flew *into* the house.
 e. Mary did not give *John a book*. *Instead*, she gave *Bill a pen*.

Condition of anaphoric *instead* more precisely:

Preceding sentence *S'* must entail or implicate the negation of what *S* describes applied to the replacee rather than the replacer.

2. when not associating with focus:

Instead requires the preceding sentence to entail or implicate the negation of *S*:

- (13) a. It should have rained. *Instead*, it snowed.
 b. People did not dance in the street. *Instead*, there was silence everywhere.
 c. John did not praise himself. *Instead*, Joe criticized Mary.
-

3. *Place*-NPs

Place-phrases can act like *instead*-phrases, but come with a counterfactual conditional force:

- (14) a. In your place, I would have declined the offer.
 b. ??In your place, the company would have hired Bill.

Place-NPs make reference to a slot in a contextually given, counterfactual situation.

Connection between antecedent and consequent must be established on the basis of empathy or simulation (putting oneself into the shoes of another, e.g. the addressee).

Place-NPs are equivalent to the antecedent of empathetic conditionals:

- (15) If I were you, I would have declined the offer.

‘You’ refers to the addressee as someone in a particular slot in the relevant contextually given situation

(14a) makes overt reference to a slot in a situation (occupied by the addressee)

(15) makes implicit reference to a slot in a situation.

4. Toward a unified semantics of replacement constructions

PLACE as a light noun -- manifestations as *place*, *stead*, *Stelle*, *statt*, *lieu*, ...

- occurs explicitly or implicitly in all replacement constructions
- stands for slots within a structured whole, on its derivative meaning (originally just a subregion of a given region)

PART: also stands for a slot – as part of a whole to which explicit reference is made.

PLACE stands for a slot, as part of a background whole / region.

A slot-based semantics of replacement constructions:

Structured whole with slots in the case of *replace*

A contextually given entity (planned event, organization, artefact, region).

Structured whole with slots in the case of *instead*

The structured proposition expressed by the sentence (that is modified) or a contextually given situation.

In all cases, the slot of the replacee and replacer is defined in terms of functions within the whole, e.g., a thematic role for an event or relations that make up a structured proposition or situation.

Using slot mereology (Bennett 2013)

Events, institutions, artefacts, regions, contextually given situations, and structured propositions come with slots

Part relation relativized to a slot, as well as a time and a world:

For an entity x and a slot s of a whole y , ' $P^{w,t}(s)xy$ ' means ' x is a part of y occupying s at t in w '.

The semantics of replacement expressions (sketch):

The verb *replace*

- (16) For objects d and d' , an event e , a structured whole b with a slot s occupied by d at the initial time of e , for an agent a ,

if $[replace\ with]^{w,t}(e, a, d, d', b) = 1$, then as a result of e , $P^{w,t}(s)d'b$.

Instead-phrases

(17) a. *Instead*-phrases with association with focus:

For a sentence S with focused constituent X , expressing a structured proposition p with a slot occupied by X ,

$$[\textit{instead of } X', S]^{w,t} = 1 \text{ iff } [S]^{w,t} = 1$$

provided $P^{w',t}(s)[X']p$ for every world w' in the common ground.

b. *Instead*-phrases without association with focus

For a contextually given situation c with a slot s for an event,

$$[\textit{instead of } S', S]^{w,t} = 1 \text{ iff there is an actual event } e \text{ that is a truthmaker of } S \text{ and}$$

$P^{w,t}(s)ec$, provided there is an event e' such that for any world w' in the common ground, $P^{w',t}(s)e'c$.

Anaphoric instead

involves anaphoric reference to the replacee.

Place-phrases

involve contextually given situations (with slots) serving in the antecedent of a conditional.

General important point about the semantics

Unified notion of a structured whole with slots applies to both

1. Ordinary objects: objects of reference in our language-independent ontology
2. Semantic objects: objects that play a role only in the context of compositional semantics, e.g., compositionally obtained structured propositions (states of affairs) or pragmatically obtained situations.

Reference

Bennett, K. (2013): 'Having a part twice over'. *The Australasian Journal of Philosophy* 91, 83-103